



Hall Ticket No-

Code: S123528

Stanley College of Engineering and Technology for Women(A)**B.E(CSE/EEE/ECE/IT/CME/AI&DS) II Semester (Backlog) Examinations May-June-2023****Mathematics-II**

Time: 3 hours

Max.Marks:60

PART-A**Note: Answer all questions (Compulsory)****5 X 2=10M**

- | | | | |
|---|--|-----|----|
| 1 | Write standard form of Bernoulli's differential equation of first order. | 2 M | L1 |
| 2 | Find the Particular Integral of $(D^2 + 4^2)y = e^{-4x}$ | 2 M | L1 |
| 3 | Define the Gamma function. | 2 M | L1 |
| 4 | State Cayley-Hamilton theorem. | 2 M | L1 |
| 5 | Find the Laplace Transform of t^3 | 2 M | L1 |

PART-B**Note: Answer all questions****5 X 10= 50M**

- | | | | |
|-----------|---|-----|----|
| 6 | a. Find the Orthogonal Trajectory of the family of curves $r^n = a^n \cos n\theta$, where θ is parameter. | 5M | L4 |
| | b. Solve $(3x^2 + 6xy^2)dx + (6x^2y + 4y^3)dy = 0$ | 5M | L3 |
| OR | | | |
| | c. Solve $(3y^2 + 4x) dx + 3xydy=0$ | 5M | L4 |
| | d. Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$ | 5M | L3 |
| 7 | a. Solve $\frac{d^2x}{dt^2} + 5 \frac{dx}{dt} + 6x = 0$, given $x(0) = 0, \frac{dx}{dt}(0) = 15$ | 5M | L3 |
| | b. Solve $(D^2 - 6D + 25)y = e^{2x}$. | 5M | L3 |
| OR | | | |
| | c. Solve by the method of variation of parameters, $\frac{d^2y}{dx^2} + 4y = \tan 2x$. | 10M | L3 |
| 8 | a. Show that $\int_a^b (x-a)^m (b-x)^n dx = (b-a)^{m+n-1} \cdot \beta(m+1, n+1)$ | 5M | L3 |
| | b. Show that $\int_0^\infty \sqrt{x} e^{-x^3} dx = \frac{\sqrt{\pi}}{3}$ | 5M | L3 |
| OR | | | |
| | c. State and Prove the relation between Beta and Gamma. | 5M | L3 |
| | d. Prove that $\beta\left(m, \frac{1}{2}\right) = 2^{2m-1} \beta(m, m)$ | 5M | L3 |

- 9 a. Reduce the quadratic form into canonical form by an orthogonal transformation. Also state the nature of quadratic form

10M L2

$$8x_1^2 + 7x_2^2 + 3x_3^2 - 12x_1x_2 - x_2x_3 + 4x_3x_1$$

OR

- c. Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} .

5M L2

- d. Find the Eigen values of $\text{adj } A$ and $A^2 - 2A + I$, where $A = \begin{pmatrix} 2 & 3 & 4 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{pmatrix}$

5M L2

- 10 a. Find the Laplace transform of $t^3 e^{-3t}$.

5M L3

- b. Find the inverse Laplace transform of the $\frac{1}{s(s^2+a^2)}$ by convolution theorem.

5M L3

OR

- c. Solve, using Laplace transform method, $y'''+2y''-y'-2y=0$ with the conditions $y(0) = y'(0) = 0$ & $y''(0)=6$

10M L3



Hall Ticket No-

Code: S123527

Stanley College of Engineering and Technology for Women(A)

B.E (CSE/CME/AI&DS/IT) II Semester (Backlog)Examinations May-June-2023

Data Structures with C

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 Explain how to find the performance of an algorithm.
- 2 Define the ADT for Deque.
- 3 Explain types of linked lists.
- 4 Describe the tree traversals with an example.
- 5 What is hashing?

2 M
2 M
2 M
2 M
2 M

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. With the help of suitable example , explain the various operations used in datastructures.

10 M

OR

- c. Write a heap sort algorithm for sorting a list of integers in ascending order.

5M

- d. Explain how linear search works with an example.

5M

- 7 a. Write a C program to implement stack using singly linked list.

5M

- b. Write a C function for inserting an integer element in to a circular queue of integers.

5M

OR

- c. Write a program to concatenate single linked lists.

5M

- d. Give brief description about the storage using arrays.

5M

- 8 a. Define Double linked list. Write an algorithm to delete an element from the existing linked

5M

- b. Discuss about linked representation of Sparse Matrix.

5M

OR

- c. Write a C program to implement stack using arrays.

5M

- d. Explain the various operations on a Single Linked List.

5M

- 9 a. Construct the binary tree of the following data 25, 30, 10, 9, 62, 5, 18, 43, 53.

5M

- b. Explain the Graph traversals with an example.

5M

OR

- c. Explain various graph representations with an example and mention the pros and cons of each representation.

5M

- d. Explain Kruskal's algorithm for obtaining the Minimal Spanning Tree of a graph.

5M

- 10 a. Write an algorithm to delete an element from the binary search tree.

5M

- b. Construct the B-Tree of order 4 of the following data 20, 30, 40, 15, 18, 16, 50, 25, 9,17,10

5M

OR

- c. Explain the various operations on AVL tree.

5M

- d. Explain Collision Resolution Methods.

5M

Hall Ticket No-

Code:S123524

Stanley College of Engineering and Technology for Women(A)

B.E(IT) II Semester (Backlog) Examinations May-June-2023

Applied Physics

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (compulsory)

5 X 2=10M

- | | | | |
|---|---|-----|----|
| 1 | Compare the properties of Laser light with ordinary light | 2 M | L1 |
| 2 | Distinguish between polar and non-polar dielectrics | 2 M | L2 |
| 3 | State the physical significance of a wave function | 2 M | L1 |
| 4 | Narrate the basic principle of working of a Photocell | 2 M | L1 |
| 5 | What is a thermistor and mention its applications | 2 M | L1 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | | | |
|---|---|----|----|
| 6 | a. Define spontaneous and stimulated emissions? Obtain expressions for relationship between the Einstein coefficients A and B. | 7M | L3 |
| | b. List industrial and scientific applications of Laser. | 3M | L2 |
| | OR | | |
| | c. What is an optical fibre? Demonstrate the fibre drawing process by Double crucible method. | 6M | L3 |
| | d. The refractive indices of core and cladding of an optical fibre respectively are 1.48 and 1.41. Calculate (a) fractional change in index (b) NA (c) acceptance angle and (d) critical angle. | 4M | L2 |
| 7 | a. Examine the variation of dielectric polarization with temperature and frequency and represent it graphically. | 6M | L4 |
| | b. Calculate the electronic polarizability of neon given that diameter of neon atom is 0.316 nm and $\epsilon_0 = 8.854 \times 10^{-12}$ F/m | 4M | L3 |
| | OR | | |
| | c. State the characteristics of Ferro electric materials? Discuss ferroelectricity in Barium Titanate in detail. | 7M | L5 |
| | d. Indicate any three applications of soft and hard magnetic materials. | 3M | L2 |
| 8 | a. State Meissner's effect. Formulate that zero resistivity and expulsion of magnetic field from the interior of a bulk superconductor are two distinct properties of any superconductor. | 7M | L5 |

- b. Write summary of BCS theory of superconductivity 3M L2
- OR**
- c. State de Broglie Hypothesis. Develop Schrodinger time dependent wave equations of a particle with K.E ' $p^2/2m$ ' and potential energy ' V '. 7M L6
- d. Find the first three energy levels of an electron confined to a box of width $2.25\text{\AA}$. 3M L2
- 9 a. Outline the salient features of Classical free electron theory and compile its demerits. 6M L4
- b. Give the conclusions of Kronig Penney Model 4M L2
- OR**
- c. State Hall effect? Determine expressions for Hall coefficient and Hall voltage for a p-type semiconductor sample. 7M L4
- d. Discuss formation of PN junction diode with appropriate diagrams 3M L2
- 10 a. Narrate the principle, working and construction of a solar cell. 6M L3
- b. Indicate merits and demerits of bulk materials compared to thin films. 4M L2
- OR**
- c. Explain sol-gel process to prepare nano materials. 6M L3
- d. What are properties of carbon nanotubes? State applications of CNTs. 4M L2
